Use-Case Flow Specification

Problem Statement #28 – Inter-City Freight Load Matching Marketplace

Use Case ID	UC-05
Use Case Name	Submit Electronic Proof of Delivery
Primary Actor	Freight Carrier (driver on the assigned trip)
Secondary Actors	Escrow Payment Gateway (external), Shipper (notified party)
Related Use Cases	«include» UC-07 Release Escrow Milestone Payment; «extend» UC-09 Raise Delivery Dispute
Trigger	The driver completes unloading at the consignee's site and taps "Complete Delivery" in the carrier app.
Traces To	FR-004 (POD capture and validation), NFR-001 (escrow release within 60 s)

Preconditions

1. The carrier is verified on the platform (valid permits, insurance and KYC) and is the awarded carrier for the load.
2. The shipment status is "In-Transit" and the vehicle has reached the registered drop-off location.
3. The shipper's funds for the delivery milestone are already held in escrow against the freight contract.
4. The driver is authenticated in the carrier app and the device has location services enabled.

Postconditions

1. A verified, immutable POD record is stored and linked to the Load ID and the freight contract.
2. The shipment milestone is set to "Delivered" and the load is closed.
3. The escrowed milestone amount is disbursed to the carrier within 60 seconds of POD verification, and the transaction is written to the audit log.
4. Both the shipper and the carrier hold a settlement record for the completed shipment.

Main Success Scenario

1. The Freight Carrier opens the awarded trip in the carrier app and selects "Complete Delivery". The system displays the proof-of-delivery form for that Load ID.
2. The system captures the device GPS coordinates and the current server timestamp, and pre-fills the consignee details from the freight contract.
3. The carrier scans or photographs the signed delivery note and enters the delivered quantity and any remarks.
4. The consignee signs on the device screen; the system records the e-signature together with the signatory's name.
5. The carrier submits the POD package. The system validates it: signature present, delivery note image readable, delivered quantity within the contracted tolerance, and GPS position within 500 m of the registered drop-off point.
6. On successful validation, the system stores the POD as an immutable, hash-stamped record and sets the shipment milestone to "Delivered".
7. The system notifies the Shipper that a verified POD has been filed and starts the contractual review timer.
8. The system invokes «include» UC-07 Release Escrow Milestone Payment: the escrow hold for the delivered milestone is instructed for disbursement to the carrier's registered account.
9. The Escrow Payment Gateway confirms the transfer; the system records the transaction reference in the audit log and marks the milestone as "Paid" within 60 seconds of POD verification (NFR-001).
10. Both parties receive a settlement notification and the load is closed. The use case ends successfully.

Alternate Flow 5a – POD Validation Fails

5a1. The system rejects the submission and displays the specific reason (for example, "You are 3.2 km from the drop-off point" or "Consignee signature missing").

5a2. The carrier corrects the input — moves to the correct location, re-captures the signature or re-scans the delivery note — and resubmits.

5a3. If validation fails three times, the system flags the delivery for manual verification, notifies the shipper's operations team and keeps the escrow amount on hold. The main flow resumes at step 6 only after a manual override is recorded.

Alternate Flow 7a – Shipper Disputes the Delivery

7a1. The shipper submits a dispute reason and supporting evidence (for example, short delivery or damaged cargo) within the review window; this invokes «extend» UC-09 Raise Delivery Dispute.

7a2. The system suspends the automatic disbursement, holds the escrow amount and sets the milestone status to "Disputed".

7a3. Both parties are notified and the dispute is routed for resolution. Steps 8-10 run only if the dispute is resolved in the carrier's favour; otherwise the escrow amount is refunded to the shipper.